Tong (Tony) Liu (he/him)

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Research Interests Last updated: 2026-09-12

Machine learning methods for observational causal inference. Data science and causal inference applications in healthcare, mental wellness, and social science.

Academic Appointments

Mount Holyoke College 2024 — present
Assistant Professor of Computer Science South Hadley, MA

Education

University of Pennsylvania 2018 — 2024
Ph.D. in Computer and Information Science Philadelphia, PA
Advisors: Lyle Ungar and Konrad Kording

Williams College 2012 — 2016
B.A. in Computer Science, concentration in Cognitive Science Williamstown, MA
summa cum laude, with highest honors in Computer Science

Research Areas

Machine Learning Methods for Observational Causal Inference
Developing machine learning methodology for improving efficiency of observational causal studies.
- Works: 17, 12, 7

Data Science for Social Communication
Using causal inference and interpretable ML methods to study computer-mediated social communication.
- Works: 26, 18, 1

Automated and Interpretable Machine Learning for Medicine
Developing workflows using automated machine learning methods and post-hoc interpretability to produce robust predictions for surgical hernia risk, liquid biopsy, and suicide risk.
- Works: 28, 27, 25, 24, 22, 3, 13, 8, 5, 2

Mobile Sensing for Depression and Anxiety
Analyzing digital phenotyping signals from mobile sensor data as markers for depression and anxiety.
- Works: 19, 16, 10, 11, 6

Works: Conference Proceedings (C), Journal Articles (J), Workshop Papers (W), Abstracts (A)

* indicates shared first author; † indicates Mount Holyoke College student co-author.

28 (J) Steven C. Marcus, Mark Olfson, Sara Wiesel Cullen, Nicholas Cardamone, Rebecca Del Rossi, Raina M. Merchant, **Tony Liu**, Timothy Schmutte, Ming Xie, and Nathaniel J. Williams. A Randomized Experiment of Mailed Incentives to Improve Emergency Physician Survey Participation. *Academic Emergency Medicine*, 33(7):e70385, July 2026

27 (J) Steven C. Marcus, Nathaniel J. Williams, Timothy Schmutte, Ming Xie, Sara Wiesel Cullen, **Tony Liu**, Lyle H. Ungar, Nick Cardamone, and Mark Olfson. Machine-Learning Identified Suicide Risk and Emergency Department Inpatient Admission. *General hospital psychiatry*, 101:128–131, 2026

26 (C) **Tony Liu**, Bhargavi Patil†, Thu Ngo†, Chris Karr, Theresa Nguyen, Rachel Kornfield, and Jonah Meyerhoff. Framing helper therapy to support user engagement: Causal evidence from a public deployment of a mental health support text messaging program. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, pages 1–17, 2026

25 (J) Steven C Marcus, Sara Wiesel Cullen, Ming Xie, **Tony Liu**, Nathaniel J Williams, Timothy Schmutte, Lyle H Ungar, Nicholas C Cardamone, and Mark Olfson. Comparing risk prediction for suicide attempts and deaths after emergency department visits for individuals with mental health disorders. *The Journal of clinical psychiatry*, 87(2):25m16070, 2026

24 (J) Timothy Schmutte, Steven C Marcus, Ming Xie, Sara Wiesel Cullen, **Tony Liu**, Lyle H Ungar, Nick Cardamone, Nathaniel J Williams, and Mark Olfson. Comparing patterns of recent mental health service use for predicting suicidal events following emergency department mental health visits in the united states: A national cohort study. *Social Psychiatry and Psychiatric Epidemiology*, 61(3):535–543, 2026

23 (J) Susanna Howard*, **Tony Liu***, Jesse Hsu, Daniel Zhang, John Farrar, Christina Jackson, and Stephen Bagley. Rate of biopsy prior to resection among patients with high-grade glioma: A nationwide database analysis. *Journal of Neurosurgery*, 2025

22 (J) Steven C Marcus, Sara Wiesel Cullen, Timothy Schmutte, Ming Xie, **Tony Liu**, Lyle H Ungar, Nick Cardamone, Nathaniel J Williams, and Mark Olfson. A cohort study of predictors of short-term nonfatal suicidal and self-harm events among individuals with mental health disorders treated in the emergency department. *Journal of Psychiatric Research*, 2025

21 (J) Nicholas C Cardamone, Mark Olfson, Timothy Schmutte, Lyle Ungar, **Tony Liu**, Sara W Cullen, Nathaniel J Williams, and Steven C Marcus. Classifying unstructured text in electronic health records for mental health prediction models: Large language model evaluation study. *JMIR Medical Informatics*, 2025

20 (J) Ben Baker, **Tony Liu**, Jordan Matelsky, Felipe Parodi, John W Krakauer, Brett Mensh, and Konrad Kording. Computational kinematics of dance: Distinguishing hip hop genres. *Frontiers in Robotics and AI*, 2024

19 (J) Caitlin A Stamatis, Jonah Meyerhoff, Yixuan Meng, Zhi Chong Chris Lin, Young Min Cho, **Tony Liu**, Chris J Karr, Tingting Liu, Brenda L Curtis, Lyle H Ungar, et al. Differential temporal utility of passively sensed smartphone features for depression and anxiety symptom prediction: a longitudinal cohort study. *npj Mental Health Research*, 3(1):1, 2024

18 (C) **Tony Liu**, Lyle Ungar, Konrad Kording, and Morgan McGuire. Measuring causal effects of civility without randomization. *International AAAI Conference on Web and Social Media (ICWSM)*, 2024

17 (J) **Tony Liu**, Patrick Lawlor, Lyle Ungar, Konrad Kording, and Rahul Ladhania. Automated detection of causal inference opportunities: Regression discontinuity subgroup discovery. *Transactions on Machine Learning Research (TMLR)*, 2023

16 (J) Jonah Meyerhoff, Tingting Liu, Caitlin Stamatis, **Tony Liu**, Harry Wang, Yixuan Meng, Brenda Curtis, Chris J Karr, Garrick Sherman, Lyle H Ungar, and David C Mohr. Analyzing text message linguistic features: Do people with depression communicate differently with their close and non-close contacts? *Behavior Research and Therapy*, 2023

15 (W) **Tony Liu**, Patrick Lawlor, Lyle Ungar, Konrad Kording, and Rahul Ladhania. Automated detection of interpretable causal inference opportunities: Regression discontinuity subgroup discovery. *ICML 2023 Workshop: Interpretable Machine Learning for Healthcare*, 2023

14 (P) Jordan K. Matelsky, Felipe Parodi, **Tony Liu**, Richard D. Lange, and Konrad P. Kording. A large language model-assisted education tool to provide feedback on open-ended responses. *arXiv*, 2023

13 (A) C Amro, A Desai, P Dattatri, **Tony Liu**, JY Hsu, RB Broach, LH Ungar, and JP Fischer. Leveraging natural language processing and artificial intelligence to label unstructured data for risk prediction. *British Journal of Surgery*, 2023

12 (C) **Tony Liu**, Patrick Lawlor, Lyle Ungar, and Konrad Kording. Data-driven exclusion criteria for instrumental variable studies. *Conference on causal learning and reasoning (CLeaR)*, pages 485–508, 2022

11 (J) **Tony Liu**, Jonah Meyerhoff, Johannes C Eichstaedt, Chris J Karr, Susan M Kaiser, Konrad P Kording, David C Mohr, and Lyle H Ungar. The relationship between text message sentiment and self-reported depression. *Journal of affective disorders*, 302:7–14, 2022

10 (J) Caitlin A Stamatis, Jonah Meyerhoff, Tingting Liu, Garrick Sherman, Harry Wang, **Tony Liu**, Brenda Curtis, Lyle H Ungar, and David C Mohr. Prospective associations of text-message-based sentiment with symptoms of depression, generalized anxiety, and social anxiety. *Depression and anxiety*, 2022

9 (J) Bernard t Hart, Titipat Achakulvisut, Ayoade Adeyemi, Athena Akrami, Bradly Alicea, Alicia Alonso-Andres, Diego Alzate-Correa, Arash Ash, Jesus Ballesteros, Aishwarya Balwani, . . . , **Tony Liu**, et al. Neuromatch academy: a 3-week, online summer school in computational neuroscience. *Journal of Open Source Education*, 5(49):118, 2022

8 (A) Ankoor A Talwar, Abhishek A Desai, Phoebe B McAuliffe, **Tony Liu**, Vivek James, Ivona Percec, Robyn B Broach, Lyle Ungar, and John P Fischer. Automated machine learning for risk prediction of incisional hernia in abdominal surgery patients. *Plastic and Reconstructive Surgery–Global Open*, 2022

7 (J) **Tony Liu**, Lyle Ungar, and Konrad Kording. Quantifying causality in data science with quasi-experiments. *Nature computational science*, 1(1):24–32, 2021

6 (J) Jonah Meyerhoff, **Tony Liu**, Konrad P Kording, Lyle H Ungar, Susan M Kaiser, Chris J Karr, and David C Mohr. Evaluation of changes in depression, anxiety, and social anxiety using smartphone sensor features: longitudinal cohort study. *Journal of medical Internet research*, 23(9):e22844, 2021

5 (J) Omar Elfanagely, Yoshiko Toyoda, Sammy Othman, Joseph A Mellia, Marten Basta, **Tony Liu**, Konrad Kording, Lyle Ungar, and John P Fischer. Machine learning and surgical outcomes prediction: a systematic review. *Journal of Surgical Research*, 264:346–361, 2021

4 (W) **Tony Liu**, Patrick Lawlor, Lyle Ungar, and Konrad Kording. Data-driven exclusion criteria for instrumental variable studies. *ICML 2021 Workshop: Neglected Asumptions of Causal Inference*, 2021

3 (W) **Tony Liu** and Lyle Ungar. Towards cotenable and causal shapley feature explanations. *AAAI 2021 Workshop: Trustworthy AI for Healthcare*, 2021

2 (J) Hanfei Shen, **Tony Liu**, Jesse Cui, Piyush Borole, Ari Benjamin, Konrad Kording, and David Issadore. A web-based automated machine learning platform to analyze liquid biopsy data. *Lab on a Chip*, 20(12):2166–2174, 2020

1 (C) **Tony Liu**, Jennifer Nicholas, Max M Theilig, Sharath C Guntuku, Konrad Kording, David C Mohr, and Lyle Ungar. Machine learning for phone-based relationship estimation: the need to consider population heterogeneity. *Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies (IMWUT)*, 3(4):1–23, 2019

Invited Talks

Northwestern University Center for Behavioral Intervention Technologies
Causality and Machine Learning for Digital Mental Health — March 2026

Teaching

Mount Holyoke College
COMSC 341CD: Causal Inference for Data Science — S25, F25, F26
COMSC 335: Machine Learning — S26
COMSC 205: Data Structures — F24, S25, F25, S26, F26

University of Pennsylvania
CIS 1920: Python Programming — F20, F21, S23

Independent Studies

[Honors Work] Tabular Foundation Models for Instrumental Variable Estimation — F26
Bhargavi Patil

Understanding Physician Preferences in Machine Learning Tools for Mental Health Risk — F26
Sneha Timsina

Predicting and Modeling User Disengagement from a Digital Mental Health Tool — F26
Maeve Condron

Causal Evaluation of Clinical Screening Guidelines — S26
Bhargavi Patil

Disengagement Prediction in a Digital Mental Health Tool — S26
Khanh Dinh

Causal Inference and Data Engineering for Evaluating a Mental Health Support System — S25, F25
Thu Ngo and Bhargavi Patil

Topics in Causal Inference — F24
Bhargavi Patil

Student Theses

Causal-Enhanced Spatio-Temporal Graph Neural Point Processes for Traffic Congestion Event Prediction — S26
Yifei Chen, committee member

College and Departmental Service

Data Science Committee — F25, S26, F26

Data Science Tenure-Track Search Committee — F25

Professional Service

Reviewer
AAAI AI4BC 2021, AAAI AI4BC 2022, SIGKDD 2023

Organizing Committee Member — Winter 2021
AAAI AI for Behavior Change (AI4BC) Workshop

Industry Research

Senior Research Scientist / Senior Research Program Manager — April 2022 — June 2024
Roblox (part-time)

- Continued research on measuring causal effects of civility and safety on platform [18]
- Coordinated external communication, grant administration, conference presence, and tech transfer across Roblox Research

Graduate Research Intern — October 2021 — April 2022
Roblox

- Developed observational causal methods for estimating the impact of civility on engagement [18]

Other Industry Experience

Advisory Software Engineer — August 2016 — August 2018
IBM Watson Health

Honors and Awards

Max Mintz TA Hall of Fame, Honorable Mention May 2024
University of Pennsylvania

Michigan Institute for Data Science (MIDAS) Future Leaders Summit April 2024
University of Michigan

Graduate Student Fellowship for Teaching Excellence Fall 2021 — Spring 2022
Center for Teaching and Learning, University of Pennsylvania

Travel Grant Spring 2022
SIGCSE New and Aspiring Educators Professional Development Session

Certifications

Certificate of Course Completion May 2024
Inclusive STEM Teaching Project

Certificate in College and University Teaching January 2021
Center for Teaching and Learning, University of Pennsylvania